FAMILY NAME: ....
OTHER NAME(S): ....
STUDENT NUMBER: ....
SIGNATURE: ....

UNSW, SYDNEY

SCHOOL OF MATHEMATICS AND STATISTICS

SUPPLEMENTARY EXAMINATION

Term 1, 2024

MATH5975
Introduction to Stochastic Analysis

(1) TIME ALLOWED – 2 Hours

(2) TOTAL NUMBER OF QUESTIONS – 5

(3) ANSWER ALL QUESTIONS

(4) READING TIME – 10 Minutes

(5) THE QUESTIONS ARE **NOT** OF EQUAL VALUE

(6) TOTAL NUMBER OF MARKS – 60

(7) START EACH QUESTION ON A NEW PAGE.

(8) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE

(9) **ONLY** CALCULATORS WITH AN AFFIXED "UNSW APPROVED" STICKER MAY BE USED

(10) THIS EXAMINATION PAPER HAS 4 PAGES

(11) THE FOLLOWING MATERIALS WILL BE PROVIDED: Table of Probability Distributions

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. **[9 marks]** Consider the probability space $(\Omega, \mathcal{F}) = ([0,1], \mathcal{B}([0,1]))$ and the probability measure $\mathbb{P} = \lambda$, where λ is the Lebesgue measure. That is $\lambda([a,b]) = b - a$ for any subinterval $[a,b]$ of $[0,1]$. Consider the functions

$$X(\omega) = \omega - \frac{1}{2}, \qquad \omega \in [0,1]$$

and $Y = \text{sign}(X)$, that is

$$Y(\omega) = \begin{cases} 1, & \text{if } X > 0 \\ -1, & \text{if } X \leq 0. \end{cases}$$

i) Write down the σ-algebra $\sigma(Y)$.

ii) Find the cumulative distribution functions of X.

iii) Compute the expected value of X that is $\mathbb{E}(X)$.

iv) Find $\mathbb{E}(X|Y)$ as function of the variable ω.

2. **[18 marks]** Let W be a standard Brownian motion and consider for $t \in [0,T]$ the process

$$X_t = \int_0^t \cos(s) dW_s.$$

i) Compute the expected value and covariance function of the process X. That is $\mathbb{E}(X_t)$ and $\text{Cov}(X_t, X_s)$ for $s, t > 0$.

ii) Show that X admits the following representation

$$X_t = W_t \cos(t) + \int_0^t W_u \sin(u) du.$$

iii) Show that the quadratic variation of X is given by

$$\langle X \rangle_t = \frac{1}{2}\left(t + \frac{1}{2}\sin(2t)\right).$$

Hint: $\cos(2x) = 1 - 2\sin^2(x)$.

iv) Find the predictable process $\gamma \in \mathcal{L}^2_W = \{\gamma : \mathbb{E}[\int_0^T \gamma_s^2 ds] < \infty\}$ such that

$$\mathbb{E}[X_T^2|\mathcal{F}_t] = \mathbb{E}[X_T^2] + \int_0^t \gamma_s dW_s.$$

v) Find the function $v(t,x)$ such that

$$\mathbb{E}[X_T^2|\mathcal{F}_t] = v(t, X_t),$$

and show that $v(t,x)$ satisfies the following PDE.

$$\begin{cases} \partial_t v(t,x) + \frac{1}{2}\cos^2(s)\partial_x^2 v(t,x) = 0, \\ u(T,x) = x^2. \end{cases}$$

3. [**9 marks**] Suppose X and Y are continuous semimartingales such that $X_0 = 1$ and $Y_0 = 1$, and recall that the stochastic exponential of X is given by

$$\mathcal{E}_t(X) = \exp\left(X_t - \frac{1}{2}\langle X\rangle_t\right).$$

i) Show that $\mathcal{E}_t(X)\mathcal{E}_t(Y) = \mathcal{E}_t(X + Y + \langle X, Y\rangle)$

ii) Let $X_t = W_t$ and $Y_t = \lambda t$ where $\lambda > 0$.

a) Compute the semimartingale decomposition of $Z_t = \mathcal{E}_t(X + Y)$ and show that it is a submartingale.

b) Show that $Z_t = \mathcal{E}_t(X + Y)$ exhibits a multiplicative representation. In other words, find the martingale M and the increasing process A such that $Z_t = M_t A_t$.

4. [**9 marks**] Consider the stochastic differential equation

$$\begin{aligned} dX_t &= \frac{1}{2}\tanh\left(\frac{X_t}{2}\right)dt + dW_t, \quad t \in [0, \infty), \\ X_0 &= x, \end{aligned} \tag{1}$$

where W is a standard Brownian motion and we recall the following facts

- $\tanh(x) = \frac{e^{2x}-1}{e^{2x}+1}$ and $\mathrm{sech}(x) = \frac{2e^x}{e^{2x}+1}$,
- $\partial_x \tanh(x) = \mathrm{sech}^2(x)$,
- $\partial_x \mathrm{sech}(x) = -\mathrm{sech}(x)\tanh(x)$.

i) Show that these exists a unique strong solution to the SDE (1).

ii) Let f be the standard logit function given by $f(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x}$. Show that $Y_t = f(X_t)$ satisfies the SDE

$$\begin{aligned} dY_t &= Y_t(1 - Y_t)dW_t, \\ Y_0 &= f(x). \end{aligned}$$

5. [**15 marks**] Let $\mathbb{F} = (\mathcal{F}_t)_{t\le T}$ be a given filtration. For $t \in [0, T]$, let

$$L_t = e^{-\frac{1}{2}\gamma^2 t + \gamma W_t},$$

where W is a standard Brownian motion. Define a probability $\mathbb{Q}$ on $(\Omega, \mathcal{F}_T)$ through the Radon–Nikodym density $\frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_T} = L_T$.

i) Let $X_t = W_t - \gamma t$ and show that $\mathbb{E}_{\mathbb{Q}}[X_t] = 0$ and $\mathbb{E}_{\mathbb{Q}}[X_t^2] = t$.

ii) Show that the following equality holds

$$\mathbb{E}_{\mathbb{P}}(L_T \ln L_T) = \frac{1}{2}\gamma^2 T.$$

iii) Deduce that if $Z \sim N(0, \sigma^2)$, then $\mathbb{E}_{\mathbb{P}}(Ze^Z) = \sigma^2 \exp\left(\frac{1}{2}\sigma^2\right)$.

END OF EXAMINATION

X	mass/density function	domain	mean	variance	*mgf* $m_X(u) = E(e^{uX})$
Bernoulli	$P(X=1)=p$ $P(X=0)=q=1-p$	$\{0,1\}$	p	pq	$q+pe^u$
Binomial $Bin(n,p)$	$\binom{n}{k}p^k(1-p)^{n-k}$	$k=0,1,\ldots,n$	np	$np(1-p)$	$(1-p+pe^u)^n$
Geometric	$p(1-p)^{k-1}$	$k=1,2,\ldots$	$\frac{1}{p}$	$\frac{(1-p)}{p^2}$	$\frac{p}{e^{-u}-1+p}$
Poisson	$e^{-\lambda}\lambda^k/k!$	$k=0,1,\ldots$	λ	λ	$\exp\{\lambda(e^u-1)\}$
Uniform	$(b-a)^{-1}$	$x\in(a,b)$	$\frac{1}{2}(a+b)$	$\frac{1}{12}(b-a)^2$	$\frac{e^{bu}-e^{au}}{u(b-a)}$
Exponential	$\frac{1}{\beta}e^{-\frac{1}{\beta}x}$	$x\in(0,\infty)$	β	β^2	$\frac{1}{1-\beta u}$
Normal $N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\,\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$x\in(-\infty,\infty)$	μ	σ^2	$e^{\mu u+\frac{1}{2}\sigma^2u^2}$
Gamma(α,β)	$\frac{e^{-x/\beta}x^{\alpha-1}}{\Gamma(\alpha)\beta^\alpha}$	$x\in(0,\infty)$	$\alpha\beta$	$\alpha\beta^2$	$\left(\frac{1}{1-\beta u}\right)^\alpha$
Beta(α,β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$x\in[0,1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	